

Finite-Sample Long-Memory Estimation via Detectability Boundaries

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February 2026

Abstract

We propose a finite-sample framework for removing long-range dependence in equity log-price series based on a *detectability boundary* d^{LTM} : the largest fractional differencing order at which hyperbolic ACF tail structure remains statistically distinguishable from noise. Under stationary ARFIMA dynamics, d^{LTM} satisfies $d^{\text{LTM}} \approx d^* - d_0$, where d^* is the true fractional integration order and d_0 is a finite-sample resolution limit. Monte Carlo calibration yields the scaling law for ARFIMA $\hat{d}_0(T) = 5.81 \cdot T^{-0.367}$ ($R^2 = 0.999$), with $d_0 = 0.30$ at $T = 3,200$ and $d_0 = 0.20$ at $T = 10,000$. In market data, regime heterogeneity and volatility clustering make d_0 persistence estimator instead of true memory, making d^{LTM} a reliable differencing index in practice.

Applied to the S&P 500 universe over 2010–2020, d^{LTM} is estimated on a rolling $n = 2,400$ window and evaluated monthly against a fixed $d = 0.5$ benchmark across 484 assets. d^{LTM} produces whiter AR(1) residuals in 70–90% of assets at every evaluation date over a full decade. The win rate rises monotonically from near 50% when $d^{\text{LTM}} \approx 0.5$ to 97–100% in the quartile where the two approaches diverge most, and the Wilcoxon signed-rank test rejects equality at $p < 0.0001$ at every date. Results hold broadly across sectors.

1 Introduction

Financial time series often exhibit strong persistence. In equity markets, the autocorrelation function (ACF) of log-price increments typically decays more slowly than the exponential pattern implied by short-memory processes, giving rise to long memory or long-range dependence [5, 6, 2]. For fractionally integrated processes, this persistence manifests as hyperbolic decay of the ACF at large lags.

Removing long-range dependence while preserving short-run dynamics is nontrivial. Simple first differencing often leaves residual persistence intact, while over-differencing can suppress economically relevant structure [6, 5].

Classical frequency-domain estimators, such as the local Whittle estimator and its extensions, are designed to estimate the fractional integration parameter under asymptotic assumptions [10, 11]. Empirical studies of financial returns, however, have documented substantial finite-sample instability and boundary pile-up of long-memory estimates, as well as limited ability to resolve cross-sectional heterogeneity in persistence [9, 4].

Relation to indirect inference. The proposed estimator can be viewed through the lens of *indirect inference* in the sense of Gouriéroux, Monfort, and Renault. Rather than estimating the fractional integration parameter via likelihood or frequency-domain methods, we construct a deliberately misspecified auxiliary statistic based on finite-sample tail geometry of the autocorrelation

function. A Monte Carlo–estimated binding relationship demonstrates that the structural parameter d^* is methodologically recoverable from d^{LTM} on synthetic ARFIMA data. In the market application, d^{LTM} is used directly as the differencing index: the objective is not asymptotic recovery of d^* but the operational identification of the memory–removal point under the finite–sample constraints characteristic of long–memory financial data.

1.1 Benchmark: Extended Whittle works in Monte Carlo but collapses on market data

As a reference point, we benchmark against an extended Whittle (EW) estimator on the same ARFIMA simulation design used throughout the paper. Appendix 7 shows that EW tracks d^* closely in Monte Carlo across a wide range of memory levels, with small bias and tight dispersion when the data are generated from a correctly specified ARFIMA model.

However, when applied to real equity log–price data, EW does not produce a meaningful cross–sectional distribution. As documented in Appendix 8, EW estimates are tightly clustered near the nonstationary boundary, with a median estimate of 1.0 and more than 80% of assets falling in the range $[0.98, 1.02]$. This collapse to near–unit–root behavior is inconsistent with the large empirical heterogeneity in persistence visible in the ACF tail and in regime–conditional forecast behavior.

This failure is not a contradiction with the Monte Carlo result; rather, it indicates that the frequency–domain assumptions that make EW effective under correctly specified ARFIMA dynamics are violated in real market data (finite samples, regime heterogeneity, microstructure effects, and nonstationary components). This motivates a data–adaptive, tail–geometry approach that targets an *effective* memory–removal point for forecasting rather than asymptotic identification under an idealized model. The predictive results are noted in the results section.

2 Motivation and Background

2.1 Limitations of standard persistence measures for predictive modeling

The ADF test is designed to detect a unit root and typically declares stationarity after a single difference, ignoring residual long memory. KPSS instead rejects whenever any strong persistence is present, often continuing to reject even when the relevant long memory has been removed. Neither test searches for the minimal d that kills the hyperbolic ACF tail.

Our method instead chooses the stationary point where marginal extra differencing no longer meaningfully steepens the ACF tail. That is precisely the point where long–range memory has been killed but short–run predictive structure remains.

It is also important to distinguish the proposed approach from alternative persistence diagnostics such as mutual information measures or Hurst exponent estimates. Mutual information captures general nonlinear dependence between observations but does not provide a direct notion of scale–specific decay or a criterion for removing long–range dependence via differencing. As a result, it is not naturally suited to selecting a fractional differencing order for predictive modeling.

Similarly, the Hurst exponent summarizes asymptotic scaling behavior of the process but does not directly characterize the finite–sample geometry of the ACF tail or identify the point at which hyperbolic dependence becomes statistically indistinguishable from noise. In finite samples, Hurst estimates can remain elevated even when further differencing no longer alters the effective tail structure relevant for forecasting.

By contrast, the proposed framework identifies the finite–sample detectability boundary d^{LTM} at which hyperbolic ACF tail structure is most distinctly resolved. This boundary is used directly as the

differencing index in market applications. A separate Monte Carlo calibration mapping demonstrates that the structural fractional integration order d^* is recoverable from d^{LTM} on synthetic ARFIMA data, validating the theoretical properties of the procedure. Unlike ADF or KPSS, which provide binary or global persistence diagnostics, the proposed approach directly targets the hyperbolic decay structure associated with fractional integration.

3 Background on Long Memory and Fractional Differencing

3.1 Long memory

Let $\{p_t\}_{t \geq 0}$ denote the log price of an asset, and define the log–return by $r_t = p_t - p_{t-1}$. A process exhibits *long memory* if its autocorrelations decay at a power–law rate rather than exponentially. More formally, an $I(d)$ process has fractional integration order d when its ACF satisfies

$$\rho(k) \sim C k^{2d-1} \quad \text{as } k \rightarrow \infty,$$

for some constant C . When $d > 0$, the ACF decays slowly, reflecting long–range dependence. The theoretical log–log slope of the ACF tail is linear with coefficient $2d - 1$, so estimating the slope of $\log |\rho(k)|$ against $\log k$ is a canonical way to diagnose memory.

3.2 Fractional differencing

Classical differencing operators remove integer degrees of integration. Fractional differencing extends this idea to non–integer orders: for $d \in \mathbb{R}$, the fractional difference of order d applied to the log price is

$$\Delta^d p_t = \sum_{j=0}^{J(d)} w_j(d) p_{t-j}, \tag{1}$$

where $w_j(d) = (-1)^j \binom{d}{j}$ are the fractional binomial coefficients and $J(d)$ is chosen so that terms with $|w_j(d)| < 10^{-4}$ are truncated. For $d = 1$ the operator reduces to the usual first difference, but when $0 < d < 1$ the series is fractionally integrated: the ACF decays more rapidly yet short–run structure may remain.

If p_t is an $I(d)$ process, then $\Delta^d p_t$ is stationary with short–range dependence. However, in finite samples we must calibrate d carefully: if d is chosen too small, long memory persists; if d is chosen too large, the series becomes almost white noise, sacrificing the predictive content we aim to exploit.

3.3 High–memory behavior in ARFIMA processes

For fractionally integrated processes, increasing values of the memory parameter concentrate spectral mass increasingly near zero frequency and induce extremely slow decay of autocorrelations at large lags [5, 6, 2]. When the fractional integration order d^* approaches or exceeds moderate levels (e.g. $d^* \gtrsim 0.7$), finite samples exhibit strong persistence over long horizons and behave effectively near the non–stationary boundary, despite remaining stationary in theory [7, 12].

In this regime, estimation of tail–based quantities becomes intrinsically more variable. Although long–range dependence is genuine, finite–sample realizations contain fewer effective degrees of freedom associated with low–frequency and long–lag components, leading to increased dispersion in estimated ACF slopes and related tail diagnostics [7, 8]. This behavior reflects a fundamental limitation of finite–sample inference for strongly persistent processes rather than estimator misspecification.

As a consequence, while detectability of long memory improves sharply once d^* exceeds the resolution threshold d_0 , tail-geometry diagnostics need not vary monotonically with d^* in the high-memory regime. Increased variability at high memory levels is therefore a structural feature of strongly persistent processes and motivates treating the high-memory regime separately in empirical analysis.

4 Finite-Sample Detectability Theory

4.1 Theoretical Framework

This section establishes the existence of a finite-sample resolution limit for tail-based long-memory detection and characterizes its scaling behavior via Monte Carlo simulation.

4.1.1 Setup and Notation

Consider a fractionally integrated process $\{X_t\}$ with true memory parameter $d^* \in (0, 1)$. After applying fractional differencing of order $d \in [0, d^*]$, the residual series has memory parameter

$$\Delta d := d^* - d. \quad (2)$$

The theoretical autocorrelation function of the residual process satisfies [2]

$$\rho_{\Delta d}(k) \sim C_{\Delta d} \cdot k^{2\Delta d-1} \quad \text{as } k \rightarrow \infty, \quad (3)$$

where $C_{\Delta d} > 0$ depends on Δd .

In log-log space, this hyperbolic decay becomes linear:

$$\log |\rho_{\Delta d}(k)| = \log C_{\Delta d} + (2\Delta d - 1) \log k. \quad (4)$$

4.1.2 Finite-Sample Noise Structure

For sample size T and lag k , the sample autocorrelation estimator satisfies [3]

$$\text{Var}[\hat{\rho}(k)] \approx \frac{1}{T} \left(1 + 2 \sum_{j=1}^{\infty} \rho^2(j) \right). \quad (5)$$

For long-memory processes with $\Delta d \in (0, 1/2)$, the infinite sum in (5) converges, yielding

$$\text{SD}[\hat{\rho}(k)] = \frac{\sigma_0(\Delta d)}{\sqrt{T}}, \quad (6)$$

where $\sigma_0(\Delta d)$ is a slowly varying function of the residual memory parameter, typically bounded by $\sigma_0 \in [1, 3]$ for $\Delta d < 1/4$.

Log-ACF Noise (Delta Method). Transforming to log space via the delta method yields

$$\text{SD}[\log |\hat{\rho}(k)|] \approx \frac{\text{SD}[\hat{\rho}(k)]}{|\rho(k)|} = \frac{\sigma_0/\sqrt{T}}{C_{\Delta d} \cdot k^{2\Delta d-1}}. \quad (7)$$

Remark 1. When Δd is small, the denominator $k^{2\Delta d-1} \approx k^{-1}$ decays slowly, making the noise-to-signal ratio in log space large at high lags. This is the fundamental source of the finite-sample resolution limit.

4.1.3 Tail Regression and Fit Quality

The tail-based procedure estimates Δd by regressing log-ACF tail values on log-lag over the window $[K_1, K_2]$:

$$\log |\hat{\rho}(k)| = a + \beta \log k + \epsilon_k, \quad k \in [K_1, K_2]. \quad (8)$$

The theoretical slope is $\beta_{\text{true}} = 2\Delta d - 1$. The coefficient of determination R^2 quantifies the proportion of variance in the log-ACF tail explained by the linear model. When Δd is sufficiently small, finite-sample noise dominates and R^2 approaches zero, rendering hyperbolic tail structure undetectable.

4.2 Existence of Detectability Threshold

Proposition 1 (Detectability Threshold). *Fix tail window $[K_1, K_2]$ with $K_2 < T$, and let $R_{\min}^2 \in (0, 1)$ denote the minimum acceptable fit quality for hyperbolic tail detection.*

For a fractionally integrated process with true memory $d^ \in (0, 1)$ and sample size T , there exists a finite-sample resolution threshold $d_0(T, K_2, K_1)$ such that:*

- (i) **Resolvable regime** ($d^* > d_0$): *The log-ACF tail admits a stable linear approximation with $R^2 > R_{\min}^2$, and the tail-geometry objective $\varepsilon(d)$ has a well-defined interior maximum at d_{LTM} .*
- (ii) **Unresolvable regime** ($d^* \leq d_0$): *The log-ACF tail is dominated by finite-sample noise, yielding $R^2 < R_{\min}^2$, and d_{LTM} is not well-defined (manifesting as boundary solutions or non-convergence).*

Furthermore, standard ACF asymptotics suggest d_0 should satisfy the scaling relationships:

$$d_0 \propto T^{-1/2}, \quad (9)$$

$$d_0 \propto f(K_2, K_1), \quad (10)$$

where $f(\cdot)$ is an increasing function of the tail window length.

Proof sketch. The result follows from noting that the signal-to-noise ratio in the log-ACF tail decreases as $\Delta d \rightarrow 0$. From (7), the noise variance in log space grows as Δd^{-2} , while the signal variance approaches a constant. The coefficient of determination R^2 therefore vanishes as Δd approaches zero, establishing existence of a threshold d_0 where $R^2(d_0) = R_{\min}^2$.

The scaling relationships (9)–(10) follow from standard ACF variance asymptotics [1, 3], which imply $\text{Var}[\hat{\rho}(k)] = O(T^{-1})$ uniformly over finite lag windows. The precise functional form of $f(K_2, K_1)$ depends on the summation structure of noise terms across the tail window and requires numerical calibration. $\square$

4.3 Monte Carlo Calibration

We characterize the detectability threshold empirically via Monte Carlo simulation on synthetic ARFIMA(0, d^* , 0) data. For each configuration, we define d_0 as the smallest value of d^* for which the estimation procedure achieves:

- Convergence rate $> 95\%$ (failure rate $< 5\%$)
- Median estimate $\hat{d}^* > 0$
- Stable tail-regression fit ($R^2 > 0.6$ in the detectability index calculation)

4.3.1 Sample Size Scaling

We simulate ARFIMA processes with varying sample sizes $T \in \{3200, 10000, 20000, 40000\}$ using 200 replications per d^* value. Table 1 reports the empirically determined thresholds.

Table 1: Detectability Threshold vs Sample Size

T	Empirical d_0	$\sqrt{10000/T}$	Ratio
3,200	0.30	1.77	0.17
10,000	0.20	1.00	0.20
20,000	0.15	0.71	0.21
40,000	0.12	0.50	0.24

Note: Empirical d_0 is the smallest d^* with failure rate below 5% over 200 replications. The third column shows $\sqrt{10000/T}$ for reference. The ratio $d_0/\sqrt{10000/T}$ increases slowly with T , indicating sublinear scaling relative to the naive $T^{-1/2}$ prediction.

The threshold decreases monotonically with sample size, consistent with improved signal resolution in larger samples. Fitting a power-law relationship $d_0 = \alpha T^\beta$ via least squares on log-transformed data yields:

$$\hat{d}_0(T) = 5.81 \cdot T^{-0.367} \quad (R^2 = 0.999, p < 0.001). \quad (11)$$

Figure 1 displays the empirical thresholds alongside the fitted power law in log-log coordinates, confirming strong linearity.

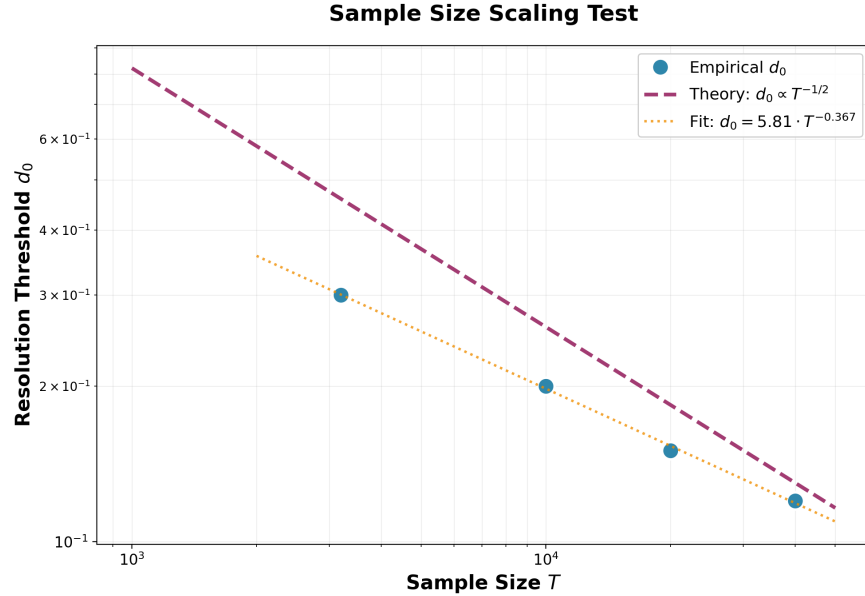


Figure 1: Sample size scaling of detectability threshold. Blue circles: empirical d_0 from Monte Carlo simulation. Red dashed line: fitted power law $\hat{d}_0 = 5.81 \cdot T^{-0.367}$ ($R^2 = 0.999$). Purple dotted line: naive $T^{-1/2}$ prediction scaled to match $d_0 = 0.20$ at $T = 10000$. The empirical scaling is significantly slower than the asymptotic prediction, indicating finite-sample effects not captured by standard ACF theory.

4.3.2 Interpretation of Empirical Exponent

The measured exponent $\hat{\beta} = -0.367$ differs from the $T^{-1/2}$ scaling suggested by standard ACF variance asymptotics (5). This discrepancy primarily reflects the finite tail window used for the log-ACF regression.

In our implementation, the regression is performed over a window $[K_1, K_2]$. The lower bound $K_1 = 20$ excludes short-lag dynamics dominated by microstructure effects and short-memory autocorrelation. The upper bound K_2 is defined as the largest lag at which the empirical autocorrelation remains statistically significant under the conventional 5% ACF significance threshold. To prevent extreme-lag noise from entering the regression, K_2 is additionally capped at 200.

Because the regression window is defined in lag space rather than scaling proportionally with the sample size T , the effective portion of the series contributing to the tail geometry does not expand linearly with T . This finite-window structure alters the signal-to-noise balance of the log-ACF regression relative to the idealized asymptotic setting, producing an empirical detectability scaling that deviates from the $T^{-1/2}$ benchmark.

While a complete finite-sample theory accounting for this effect remains to be developed, the empirical scaling law (11) provides a practical calibration formula for sample size planning. For example, to reliably detect long memory $d^{\text{LTM}} = 0.15$ with the proposed procedure, equation (11) implies a required sample size of approximately

$$T \approx \left(\frac{5.81}{0.15} \right)^{1/0.367} \approx 22,000 \text{ observations.} \quad (12)$$

4.3.3 Asymptotic Consistency

Despite the slower-than-predicted finite-sample decay, the fitted relationship (11) implies $d_0(T) \rightarrow 0$ as $T \rightarrow \infty$. Thus, arbitrarily weak long memory becomes detectable in sufficiently large samples, establishing asymptotic consistency of the tail-based procedure.

Corollary 1 (Asymptotic Detectability). *Under the empirical scaling law (11), the detectability threshold vanishes in the limit:*

$$\lim_{T \rightarrow \infty} d_0(T) = \lim_{T \rightarrow \infty} 5.81 \cdot T^{-0.367} = 0. \quad (13)$$

4.4 Practical Implications

The empirically calibrated detectability threshold has several practical consequences:

1. **Sample size requirements:** Detecting weak long memory requires substantially larger samples than suggested by asymptotic theory. The empirical exponent -0.367 implies that doubling the sample size reduces d_0 by a factor of $2^{0.367} \approx 1.29$, compared to $\sqrt{2} \approx 1.41$ under the naive $T^{-1/2}$ rule.

4.5 Comparison with Classical Estimators

The finite-sample resolution limit d_0 is specific to tail-geometry methods and does not directly apply to frequency-domain estimators such as the local Whittle or extended Whittle procedures. These estimators achieve $\sqrt{T}$ -consistency under correct model specification [10, 11], but are known to suffer from boundary pile-up and instability in empirical financial data [9, 4].

Our approach sacrifices asymptotic efficiency in favor of finite-sample robustness: rather than estimating d^* via maximum likelihood under an idealized spectral model, we target the operational

point at which long-range dependence ceases to be empirically detectable via ACF tail geometry. This shift in estimand—from a structural parameter to a detectability-based quantity—explains why the tail-based procedure produces dispersed cross-sectional estimates on equity data, while extended Whittle collapses to the nonstationary boundary (Appendix 8).

The empirical scaling law (11) formalizes this tradeoff: the detectability threshold vanishes asymptotically, but at a rate slower than the Cramér-Rao bound. For realistic sample sizes ($T < 10,000$), this imposes a nontrivial lower bound on resolvable memory strength, which we operationalize via entropy-based classification (Appendix B).

Proof sketch. The result follows from noting that the signal-to-noise ratio in the log-ACF tail decreases as $\Delta d \rightarrow 0$. From equation (7), the noise variance in log space grows as Δd^{-2} , while the signal variance approaches a constant. The coefficient of determination R^2 therefore vanishes as Δd approaches zero, establishing existence of a threshold d_0 where $R^2(d_0) = R_{\min}^2$.

The scaling relationships follow from standard ACF variance asymptotics, which imply $\text{Var}[\hat{\rho}(k)] = O(T^{-1})$. The precise functional form requires numerical calibration. $\square$

4.6 Detectability in market data relative to the ARFIMA calibration

The Monte Carlo calibration in Section 4 establishes a conservative detectability threshold $d_0(T)$ under stationary ARFIMA dynamics, fit on samples $T \in \{3200, 10000, 20000, 40000\}$. Under this null, the empirical scaling law (11) implies that reliably detecting $d_0 \approx 0.26$ requires approximately $T \approx 4800$ observations. In practice, however, the entropy-based classification procedure (Appendix B) separates genuine detectability events from convergence failures, effectively reducing the operational sample size requirement to $n = 3200$ — consistent with the convergence analysis in Section 7.

Critically, the ARFIMA calibration represents a *worst-case* benchmark for detectability. Real equity series differ fundamentally from a homogeneous fractionally integrated null: they exhibit regime heterogeneity, volatility clustering, asymmetric shocks, and structural breaks, all of which introduce additional curvature and contrast in the ACF tail. These features strengthen the effective signal-to-noise ratio of tail geometry diagnostics, implying that d_0 for market data is lower than the ARFIMA-calibrated threshold — the d^{LTM} point is easier to identify in practice than the synthetic calibration suggests.

The procedure therefore targets a *detectability boundary* rather than a pointwise parameter estimate, and this boundary-detection framing is inherently more robust to finite-sample noise than classical estimation. The Monte Carlo threshold $d_0(T)$ should accordingly be interpreted as an upper bound on the effective resolution limit in empirical market applications.

5 Methodology

5.1 Residual-based fractional differencing index d^*

Fractional differencing with order d produces

$$\Delta^d p_t = \sum_{j=0}^{J(d)} w_j(d) p_{t-j},$$

where $w_j(d)$ are binomial coefficients truncated when $|w_j(d)| < 10^{-4}$.

For each candidate $d \in \{0, 0.02, \dots, 1.0\}$, we:

1. compute the fractionally differenced series $\Delta^d p_t$;

2. estimate its autocorrelation function $\rho_d(k)$ over lags $k \in [20, 200]$;
3. approximate the tail geometry via the log–log regression

$$\log |\rho_d(k)| = a(d) + \beta(d) \log k + \varepsilon(d, k).$$

For an ideal $I(d)$ process, the theoretical ACF tail decays hyperbolically with slope $2d - 1$. In finite samples, however, both the estimated slope $\beta(d)$ and the overall tail geometry are distorted by sampling variability, microstructure effects, and regime heterogeneity. As a result, $\beta(d)$ alone is not a stable estimator of the underlying memory parameter.

Rather than targeting slope consistency, we exploit the fact that long–memory processes exhibit a distinctive *linear* structure in log–log ACF space. As fractional differencing increases, this hyperbolic structure progressively deteriorates. Once long–range dependence has been effectively removed, the remaining tail behavior becomes dominated by finite–sample noise and no longer admits a stable power–law representation.

We formalize this intuition by defining a tail–geometry deviation functional $\varepsilon(d)$, which aggregates deviations of the empirical ACF tail from its best hyperbolic approximation across the tail window. In our implementation, $\varepsilon(d)$ is the root mean squared error (RMSE) of the log–log regression of ACF tail on lag. The maximizer d^{LTM} therefore identifies the point of *maximum tail volatility* during the transition from hyperbolic decay to noise–dominated behavior, rather than the point of minimum regression error.

We then define the *long–term memory detectability boundary*

$$d^{\text{LTM}} = \arg \max_{d \in \mathcal{D}} \varepsilon(d).$$

For small d , substantial long–range dependence remains and the ACF tail preserves a coherent hyperbolic structure, yielding relatively small values of $\varepsilon(d)$. As d increases, fractional differencing attenuates this structure. Beyond a method– and sample–dependent resolution threshold, the remaining tail dependence becomes indistinguishable from noise, causing tail–geometry objectives based on $\log |\rho_d(k)|$ to deteriorate sharply. The maximizer d^{LTM} therefore marks the largest fractional differencing level at which long–memory structure remains detectably present in finite samples.

Crucially, d^{LTM} should not be interpreted as the true fractional integration order. Finite–sample noise in the tail ACF induces an empirical resolution limit, below which distinct long–memory structures cannot be reliably separated (see Monte Carlo analysis in Section 4.5 and Appendix A). Consequently, the detected boundary satisfies

$$d^{\text{LTM}} \approx d^* - d_0,$$

reflecting a detectability constraint rather than the underlying structural parameter itself.

By construction, $d^{\text{LTM}} \leq d^*$. We therefore recover the fractional differencing index d^* by applying a monotone calibration mapping to d^{LTM} . This calibration is estimated via Monte Carlo simulation using an inverse mapping that preserves order across memory regimes. Details of the calibration procedure are provided later in the paper.

Intuitively, d^* corresponds to the point at which marginal fractional differencing ceases to yield additional long–memory removal while preserving short–run dynamics relevant for forecasting.

The procedure is meaningful only when the true memory parameter lies in the detectable regime,

$$d^* \in [d_0, d_{\max}],$$

outside of which finite–sample tail diagnostics cannot reliably distinguish long–memory structure from noise.

Upper bound and high-memory instability. While the detectable region begins at d_0 , the procedure also has an effective upper limit. Empirically, as d^* enters the extreme high-memory regime, the finite-sample behavior of the detectability index deteriorates. In particular, for $d^* \gtrsim 1.0$ the distribution of d_{LTM} exhibits a pronounced increase in dispersion (with more frequent outliers and heavier tails), and the conditional median response curve $\text{median}(d_{\text{LTM}} | d^*)$ begins to lose tight tracking relative to its behavior on $[d_0, 1.0]$. This phenomenon is consistent with the well-known finite-sample difficulty of tail-based inference under extremely persistent dynamics, where effective degrees of freedom in the ACF tail collapse and estimation noise dominates local slope geometry. At present, we do not claim a sharp theoretical threshold for this breakdown: the precise dependence of this high-memory dispersion on sample size T , tail window K , and grid resolution remains to be characterized. A deeper analysis of the regime $d^* > 1.0$, including refined diagnostics and potential modifications to the objective, is left for future work. Accordingly, throughout the paper we treat d_{max} as an empirical stability bound, and interpret results beyond $d^* \approx 1.0$ with caution.

When $d^* < d_0$, the long-memory signal in the ACF tail is uniformly below the finite-sample detectability scale, and no stable interior maximizer of the tail-based objective can be expected. In this regime, the quantity d_{LTM} is not well defined in practice, and the procedure may manifest as boundary solutions or non-convergence, as documented in the Monte Carlo analysis (Section ??, Table 6).

In empirical market data, an auxiliary entropy criterion evaluated in a local neighborhood of the candidate d^{LTM} is used solely as a classification device to distinguish spurious finite-sample detections (arising when $d^* < d_0$) from genuine long-memory regimes. The entropy measure does not enter the estimation of d^{LTM} or the calibration to d^* .

Diagram below clearly shows the point d^{LTM} (0.58) for asset AES. We will discuss properties of this point in next section.

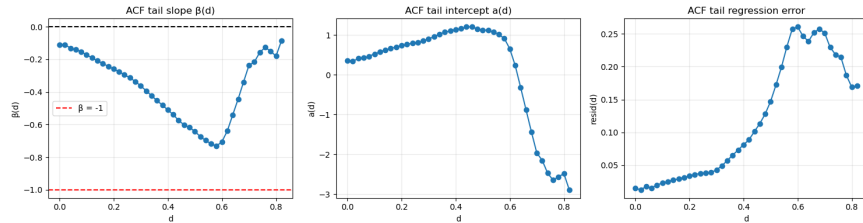


Figure 2: Regression parameter-d curve

5.2 Properties of the detectability boundary d^{LTM}

5.2.1 Sensitivity of memory to the choice of d^{LTM}

For a representative asset (AES) we plot the log-log ACF tail for:

- $d = d^{\text{LTM}} - 0.1$ (residual long-memory regime),
- $d = d^{\text{LTM}}$ (effective tail-resolution point),
- $d = d^{\text{LTM}} + 0.1$ (over-differenced regime).

Figure 3 shows log-log plots of the absolute autocorrelation tail, $\log |\rho(k)|$ versus $\log k$, for three fractional differencing orders in a neighborhood of the empirically selected d^{LTM} . For each fixed d , a linear fit is applied to the high-lag region, corresponding to a local power-law approximation $|\rho(k)| \approx Ck^{\beta(d)}$ over the chosen lag window. For values of d below this region, the tail exhibits visible

curvature, indicating that residual long-range dependence remains and that a single power-law provides a poor approximation. As d increases, the tail becomes progressively closer to linear over the selected lags, with the strongest apparent alignment occurring near d^{LTM} . Beyond this point, further differencing reduces the effective tail signal, leading to increased dispersion and instability in the fitted slope as finite-sample noise begins to dominate. Accordingly, d^{LTM} is interpreted as the differencing order at which the hyperbolic tail structure is most distinctly resolved in finite samples, rather than as a direct estimator of the true memory parameter d^* .

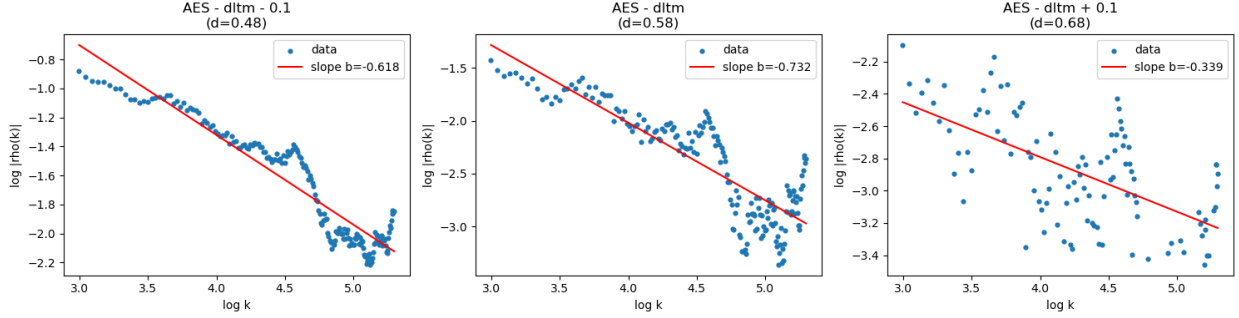


Figure 3: AES log-log ACF tail at $d^{\text{LTM}} - 0.1$, d^{LTM} , and $d^{\text{LTM}} + 0.1$

5.2.2 Entropy behavior near the detectability boundary

A second empirical property of the detectability boundary d^{LTM} concerns the geometric regularity of the ACF tail slope as a function of the fractional differencing order. This property is captured via an entropy measure defined on the evolution of the tail-slope curve.

Let $\beta(d)$ denote the estimated slope of the log-ACF tail obtained by regressing $\log |\rho_d(k)|$ on $\log k$ over a fixed lag window $\mathcal{K}$, for each d on a discrete grid $\{d_1 < d_2 < \dots < d_m\}$. Define the discrete sign process

$$s_j = \text{sign}(\beta(d_{j+1}) - \beta(d_j)), \quad j = 1, \dots, m-1,$$

with zero increments discarded. The sequence $\{s_j\}$ encodes the local directional behavior of the tail-slope curve.

We measure the irregularity of this sequence using the *sign-change entropy*. Let

$$p = \mathbb{P}(s_{j+1} \neq s_j)$$

denote the empirical probability of a sign flip between successive increments. The normalized Shannon entropy is defined as

$$H = -\frac{p \log p + (1-p) \log(1-p)}{\log 2}, \quad H \in [0, 1].$$

Low values of H correspond to a predominantly monotone slope curve, whereas values close to one indicate frequent directional reversals and lack of geometric coherence.

To assess stability in the vicinity of the detectability boundary, we compute this entropy locally over a neighborhood $[d^{\text{LTM}} - \delta, d^{\text{LTM}} + \delta]$. Empirically, when the true fractional integration order satisfies $d^* < d_0$, the tail-slope curve is dominated by finite-sample noise and exhibits high entropy, reflecting the absence of a resolvable hyperbolic structure. In contrast, when $d^* \geq d_0$, the slope curve becomes locally coherent near d^{LTM} , leading to substantially lower entropy.

This entropy measure does not enter the construction of d^{LTM} and does not alter the calibration to the structural fractional integration order d^* . It is used solely as a validation device on synthetic ARFIMA data to demonstrate that convergence failures — cases where $d^* < d_0$ and d^{LTM} is not well-defined — are separable from genuine long-memory detections using observable properties of the slope curve alone. In practice, we operationalize this via a binary classifier trained on entropy, mutual information, and Hurst-type features, which achieves near-perfect separability between false and true detectability cases on synthetic data (AUC = 0.9985; full performance metrics reported in Appendix B).

This classifier is *not* applied to market data. In the empirical application (Section 8), d^{LTM} is used directly without any classification or filtering step: the entropy analysis serves purely to establish that the theoretical failure modes of the procedure are diagnosable from series properties, not to gate market estimates.

The entropy classifier does not guarantee perfect separation and the classification boundary requires further theoretical characterization.

Monte Carlo evidence supporting this entropy behavior is reported in Appendix A.2, where Table 9 summarizes global and local slope-entropy diagnostics across values of the true fractional integration order.

5.3 Median-Based Indirect-Inference Calibration of the Memory Index

Let d^{LTM} denote the finite-sample long-term memory index produced by the proposed tail-based procedure, and let d^* denote the true fractional integration parameter of the underlying process. Monte Carlo simulations reveal that the conditional median response

$$m(d^*) := \text{median}(d^{\text{LTM}} \mid d^*)$$

is monotone on the identifiable region where d^{LTM} admits a stable interior maximizer.

To demonstrate that the structural parameter d^* is methodologically recoverable from d^{LTM} , we construct a calibration by inverting the median response curve. Specifically, on the domain where $m(d^*)$ is strictly increasing, we define the calibrated estimator

$$\hat{d}_{\text{cal}} := m^{-1}(d^{\text{LTM}}),$$

where m^{-1} denotes a monotone generalized inverse defined on the range of m . Plateaus in $m(d^*)$ are collapsed by assigning each constant segment a representative value of d^* , yielding a well-defined and order-preserving mapping.

Scope of the calibration. This calibration is estimated and evaluated entirely on synthetic ARFIMA data. Its purpose is to establish that d^{LTM} carries sufficient information to recover d^* under a controlled generative model — a methodological result that validates the theoretical framework. It is *not* applied in the market application: in the empirical analysis of Section 8, d^{LTM} is used directly as the differencing index without any calibration step. The detectability boundary itself, rather than the recovered d^* , is what matters for residual whitening.

Interpretation of calibration. Even under a hypothetical asymptotic analysis, the raw detectability boundary d_{LTM} would not be expected to converge directly to d^* , but rather to a deterministically shifted limit reflecting finite-sample tail-resolution effects. The role of the calibration mapping is precisely to invert this shift and recover the target fractional integration order d^* , confirming that the shift is systematic and correctable rather than irreducible noise.

5.4 Monte Carlo results

We next examine the finite-sample behavior of the proposed tail-based memory procedure using synthetic ARFIMA processes with varying true memory parameter d^* . The goal of this analysis is not to establish consistency in the classical sense, but to characterize the finite-sample detectability properties of d^{LTM} and to identify the region in which a calibrated estimator $\hat{d}_{cal}$ is meaningfully defined.

Consistent with the Monte Carlo calibration (??), when $d^* < d_0$ the tail-based objective does not admit a stable interior maximizer and the quantity d_{LTM} is not well-defined.

Accordingly, the calibrated estimator $\hat{d}_{cal}$ is defined only on the identifiable region $d^* \geq d_0$ and is treated as undefined outside this domain.

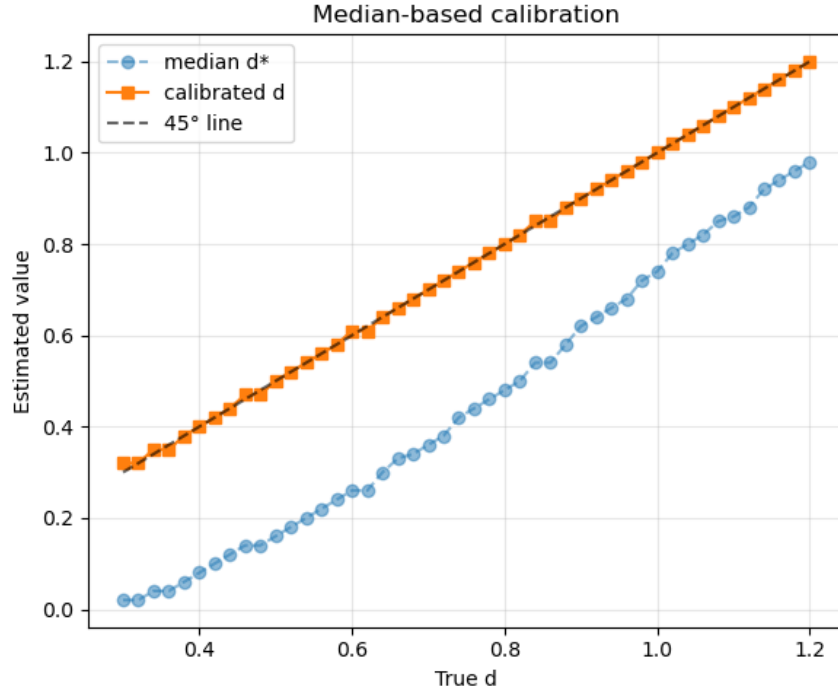


Figure 4: **Median-based calibration of the fractional differencing index.** The blue curve shows the conditional median of the detected long-term memory index d^{LTM} as a function of the true fractional integration parameter d^* , estimated from Monte Carlo simulations. The orange curve shows the calibrated estimator $\hat{d}_{cal} = m^{-1}(d^{LTM})$ obtained by inverting the median response function. The dashed line denotes the 45° identity line. Calibration is performed exclusively on the identifiable region $d^* \geq d_0 = 0.26$, below which d^{LTM} is not well-defined due to finite-sample detectability limits for n used. On this domain, the calibrated estimator achieves $R^2 = 0.9943$ with respect to the true d^* , indicating effective removal of structural bias.

This calibration procedure operationalizes the finite-sample detectability limits identified via Monte Carlo simulation by explicitly restricting estimation to the region where the long-memory signal is statistically resolvable.

Full Monte Carlo diagnostics, robustness checks, and regime-conditional performance summaries are reported in Appendix 6.

Finite-sample dispersion. An important feature of the proposed tail-based procedure is that, conditional on resolvability, it does not exhibit excessive statistical variability. As shown in Appendix Table 6, on the identifiable region $d^* \geq d_0$ the finite-sample dispersion of the detected long-term memory index d_{LTM} is comparable to that of standard semiparametric estimators such as the extended Whittle estimator under correctly specified ARFIMA dynamics. In particular, once the detectability threshold is exceeded, d_{LTM} displays tight interquantile ranges and low failure rates, indicating that the tail-geometry objective is not intrinsically noisy. The distinction between d_{LTM} and classical estimators therefore lies not in variance, but in the estimand itself: d_{LTM} targets a detectability-shifted boundary induced by finite-sample resolution limits rather than the structural memory parameter directly.

Throughout Sections 6, 7 and 8, d_{LTM} denotes the empirically estimated, walk-forward detectability boundary used in market applications. Any discrepancy in notation is purely a error, please treat as such.

6 Stability of d_{LTM} Across Estimation Windows

To assess whether d_{LTM} reflects a genuine structural property of equity log-prices rather than a sampling artefact, we re-estimate it using rolling windows of length $n \in \{800, 1200, 1600, 2000, 2400, 2800, 3200\}$ trading days across all 484 S&P 500 constituents over 2000–2025 (effective results from 2011 because 3200 n corresponds to roughly 11 years of data).

Cross-sectional distribution. Table 2 reports the cross-sectional distribution of $\hat{d}_{\text{LTM}}(n)$ at the end of the sample. Estimated d_{LTM} rises monotonically with window length, from a median of 0.06 at $n = 800$ to 0.72 at $n = 3200$. This reflects a well-documented small-sample downward bias: short windows lack sufficient power to detect hyperbolically decaying autocorrelations, causing the estimator to understate the true memory order.

Table 2: Cross-sectional distribution of $\hat{d}_{\text{LTM}}(n)$ at end of sample, 484 constituents.

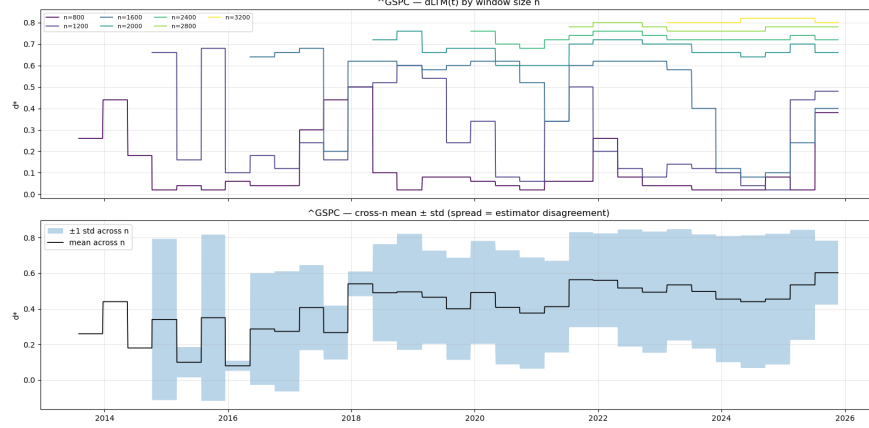
	$n=800$	$n=1200$	$n=1600$	$n=2000$	$n=2400$	$n=2800$	$n=3200$
Mean	0.114	0.219	0.313	0.414	0.542	0.620	0.673
Median	0.060	0.160	0.300	0.450	0.600	0.680	0.720
Std	0.108	0.185	0.218	0.229	0.200	0.191	0.172

Temporal stability. The time-series standard deviation of $\hat{d}_{\text{LTM}}(n, t)$, averaged cross-sectionally, follows a U-shaped pattern in n (Table 3). Instability peaks at $n \approx 1600$ (mean std = 0.194), where the window is long enough to escape the zero floor but short enough to be materially distorted by rolling through high-volatility regimes such as the 2020 COVID-19 shock. For $n \geq 2800$ the median std falls below 0.05, indicating that d^* varies slowly once estimated over a sufficiently long history.

Table 3: Mean time-series standard deviation of $\hat{d}_{\text{LTM}}(n, t)$ across tickers. Lower = more stable.

	$n=800$	$n=1200$	$n=1600$	$n=2000$	$n=2400$	$n=2800$	$n=3200$
Mean std	0.107	0.160	0.194	0.156	0.109	0.065	0.043
Median std	0.106	0.160	0.196	0.161	0.097	0.042	0.024

Bias relative to the long-window benchmark. Taking $\hat{d}_{\text{LTM}}(n)(3200)$ as the ground-truth benchmark, the mean bias $\hat{d}_{\text{LTM}}(n) - \hat{d}_{\text{LTM}}(n)(3200)$ is uniformly negative and shrinks monotonically: from -0.582 at $n = 800$ to -0.061 at $n = 2800$. Crucially, the *variance* of the bias is small for large n ($\text{std} < 0.01$ for $n \geq 2000$), meaning the bias is nearly deterministic and amenable to isotonic calibration. The correlation of $\hat{d}_{\text{LTM}}(n)(n, \cdot)$ with $\hat{d}_{\text{LTM}}(n)(3200, \cdot)$ rises from 0.04 at $n = 800$ to a median of 0.48 at $n = 2000$, confirming that a minimum of approximately 2000 observations is required for the estimator to track the true memory structure reliably.



Taken together, these results support using $\hat{d}_{\text{LTM}}(n)$ as a slowly varying but genuinely informative parameter, provided estimation is conducted over windows of at least 2800 days and the stability can be monitored as a real-time reliability flag.

Retrospective versus walk-forward estimation. The analysis in this section is retrospective in nature. Although estimates are computed on rolling windows, the objective is to study cross-window stability and convergence properties of the memory index rather than to emulate a real-time decision process. Estimates at time t are therefore used purely for diagnostic comparison across window lengths, not as inputs to a forecasting or trading rule. No claim of real-time operability is made in this section.

7 Estimator Convergence Across Window Size

Walk-forward structure of the convergence analysis. Unlike Section 6, the convergence analysis in this section is conducted in a strictly walk-forward manner. At each calendar time t , the memory index is estimated using only the most recent n observations available at that time, and the resulting sequence $\{\hat{d}(n, t)\}_t$ is analyzed for cross-window agreement and temporal stabilization as data accumulate. No future observations enter the estimation window at any point. The purpose of this walk-forward design is not forecasting, but to verify that the estimator converges to a stable quantity as sample size grows under realistic, time-ordered data availability.

A prerequisite for using d_{LTM} as a meaningful economic quantity is that the estimator converges as data accumulates. Figure 5 displays the cross-sectional distribution of d_{LTM} at the end of the sample across all seven window sizes.

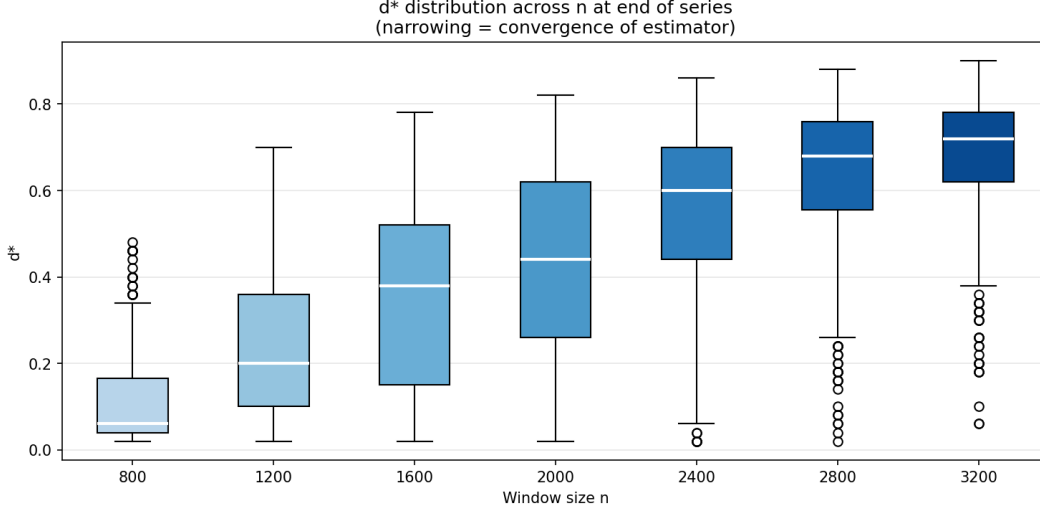


Figure 5: Cross-sectional distribution of d_{LTM} by estimation window size n at end of sample. Each box represents the interquartile range across tickers; whiskers extend to $1.5 \times \text{IQR}$. Narrowing boxes indicate the estimator settling toward a stable value as n increases.

Table 4 reports cross-sectional statistics and cross- n correlations.

Table 4: Convergence of d_{LTM} across window sizes. Correlation is between adjacent n estimates at end of series, computed on the common set of tickers.

n	Tickers	Median	Std	Q25	Q75
800	278	0.060	0.116	0.040	0.155
1200	278	0.200	0.175	0.100	0.360
1600	277	0.380	0.221	0.140	0.520
2000	273	0.460	0.230	0.260	0.640
2400	267	0.600	0.202	0.420	0.700
2800	262	0.670	0.190	0.560	0.760
3200	256	0.720	0.171	0.620	0.780

Adjacent n pair	Pearson r
$n = 800$ vs $n = 1200$	0.293
$n = 1200$ vs $n = 1600$	0.407
$n = 1600$ vs $n = 2000$	0.710
$n = 2000$ vs $n = 2400$	0.745
$n = 2400$ vs $n = 2800$	0.803
$n = 2800$ vs $n = 3200$	0.900

The correlation between adjacent estimates rises sharply from $r = 0.293$ at small n to $r = 0.900$ between $n = 2800$ and $n = 3200$, confirming convergence. Estimates at $n \leq 1200$ are dominated by estimation noise — the low correlation reflects the inability of short windows to distinguish true long memory from short-term momentum.

Remark 2. The instability of d^{LTM} at small n provides a natural, data-driven criterion for the

minimum sample size required before estimates are reliable: approximately $n \geq 2000$ trading days (≈ 8 years of daily data). We adopt $n = 2400$ as the primary window for all subsequent analyses. This length achieves high correlation with the largest window ($r = 0.803$, Table 4), materially reduced time-series variability (Table 3), and retains the broadest cross-sectional coverage among windows that have escaped the small-sample bias regime characteristic of $n < 1600$.

8 Residual Whiteness: d_{LTM} vs Fixed $d = 0.5$

8.1 Rolling Estimation of d^{LTM}

Operational distinction between d^* and d_{LTM} . In the market application, we do not use the calibrated structural parameter d^* . Instead, all empirical results in this section are based directly on the detectability boundary d_{LTM} , estimated in a walk-forward manner and applied as the fractional differencing order at decision time. This distinction is intentional: while d^* serves as a theoretical construct and calibration target under an ARFIMA null, the operational objective in market data is residual whitening under finite-sample constraints. The detectability boundary d_{LTM} is therefore the economically relevant object for real-time implementation.

The calibration and classifier results reported in Sections 5 and Appendix B serve to validate the theoretical properties of d^{LTM} on synthetic data; they are not applied to market series.

At each decision time t , we apply the tail-geometry procedure of Section 5.1 to the most recent $n = 2400$ log-price observations for asset i , yielding a rolling estimate $d_{i,t}^{\text{LTM}}$. To reduce computational cost, the estimate is recomputed every 50 trading days and held constant in between; since d^{LTM} is a slowly varying quantity (Section 7), this introduces negligible error relative to daily recomputation.

Crucially, $d_{i,t}^{\text{LTM}}$ is constructed using only data available at time t : no future prices enter the window and no look-ahead bias is possible. The differencing order applied at evaluation date t is therefore strictly predetermined.

8.2 d_{LTM} as a Time-Varying State Variable

Figure 6 plots the cross-sectional median and interquartile range of d_{2000}^{LTM} over the full sample, together with cross-sectional dispersion and the median by window size. The figure reveals that d^{LTM} is not a static parameter but a genuinely time-varying quantity that responds coherently to major macroeconomic regimes.

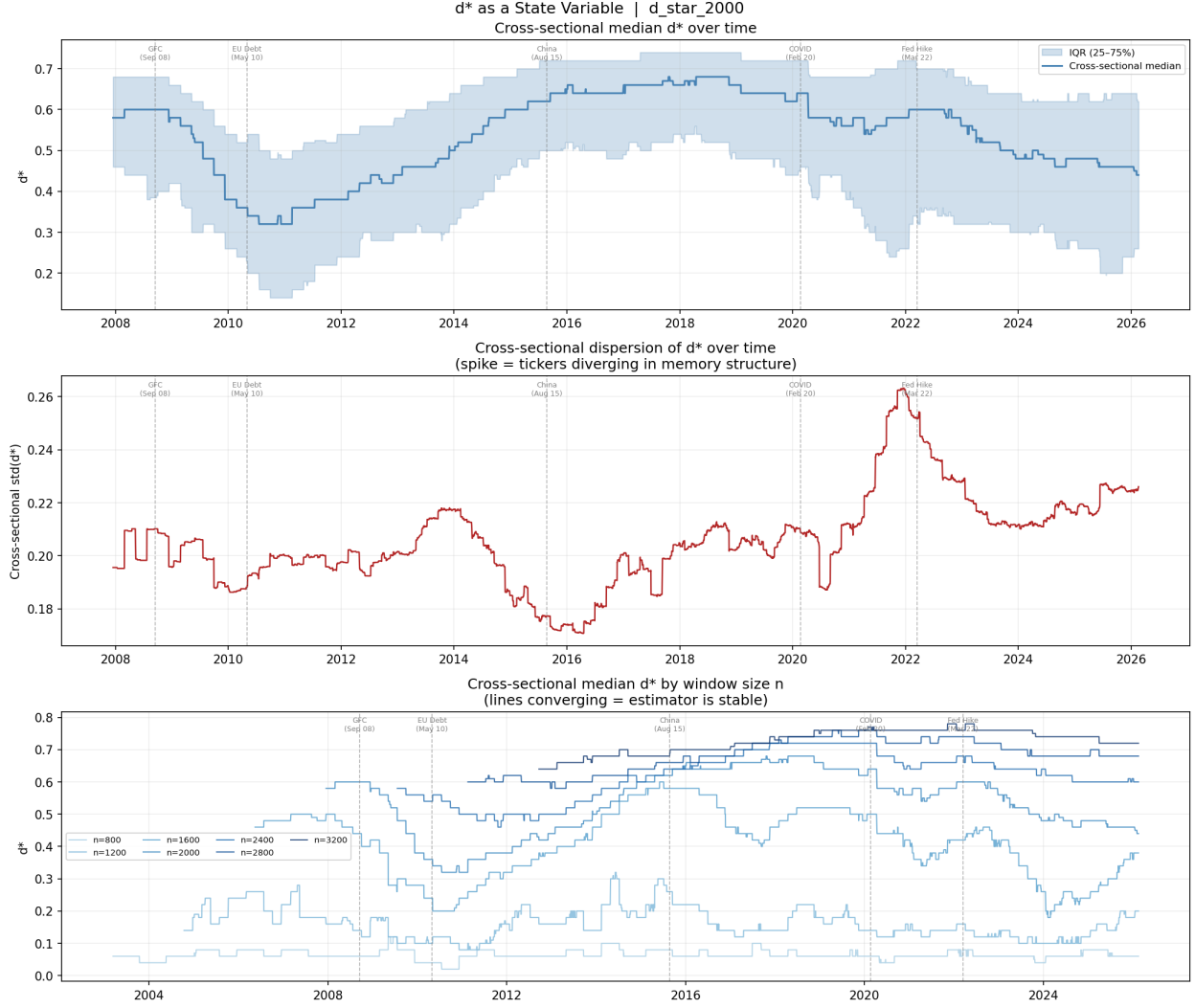


Figure 6: Top: Cross-sectional median $\hat{d}_{2000}^*$ (blue) with interquartile band. Middle: Cross-sectional standard deviation of $\hat{d}_{2000}^*$ over time. Bottom: Median $\hat{d}^*$ by estimation window size n . Vertical dashed lines mark major macroeconomic events.

The top panel tells a coherent macroeconomic story. The median d^* enters the sample elevated near 0.60 in the years preceding the Global Financial Crisis — a period of sustained, low-volatility trend-following across equities. The GFC (September 2008) triggers a sharp decline: as prices became dominated by panic-driven mean-reversion and structural breaks, the long-memory component deteriorates and the median d^* troughs near 0.32 by late 2010. The subsequent European Debt Crisis (May 2010) prolongs the trough before a slow recovery takes hold, driven by coordinated central bank intervention and the onset of quantitative easing. By 2015–2016 the median has recovered to approximately 0.55–0.60, with a brief disruption around the China growth scare (August 2015). The QE-driven bull market sustains elevated d^* through 2017–2020, peaking near 0.65 immediately before the COVID shock (February 2020). The pandemic triggers another sharp decline as cross-sectional heterogeneity spikes: the middle panel shows dispersion reaching a local maximum around the Federal Reserve hiking cycle (May 2022), indicating that tickers were diverging sharply in their memory structure as different sectors adapted at different speeds to the new rate environment. From

2022 onward, the median d^* drifts lower and stabilises near 0.45–0.50, consistent with a market that has transitioned away from the persistent, low-rate trending regime of the prior decade.

The bottom panel confirms that this time-variation is not an artefact of window-length choice: estimates across all seven values of n move in the same direction at each macro event, and converge increasingly tightly for $n \geq 2000$, consistent with the convergence analysis in Section 7.

Proposition 2. *A fixed differencing parameter $d \in (0, 1)$ is a degenerate special case of d^* in which the estimator is constrained to return the same value for all assets and all time periods. The state variable content of d^* — its time-series variation and cross-sectional heterogeneity — is eliminated by construction under the fixed- d assumption.*

8.3 Test Design

We evaluate the rolling d^{LTM} against the fixed- $d = 0.5$ benchmark monthly from September 2010 to February 2020. This window is not chosen arbitrarily: Figure 6 shows it corresponds to the largest and most persistent deviations of the cross-sectional median d^{LTM} from 0.5, bracketing a full decade of material misspecification under fixed differencing and thereby maximising the power of the comparison.

At each evaluation date we: (i) extract the most recent $n = 2400$ log-price observations for each ticker; (ii) apply fractional differencing with $d = d^{\text{LTM}}$ and $d = 0.5$; (iii) fit AR(1) to each differenced series; and (iv) apply the Ljung–Box test at lag 10 to the residuals. A higher p -value indicates whiter residuals and a better-specified model. We classify a ticker as a *win* for d^{LTM} if $p^{(d^{\text{LTM}})} > p^{(0.5)}$.

8.4 Results Across the Full Evaluation Window

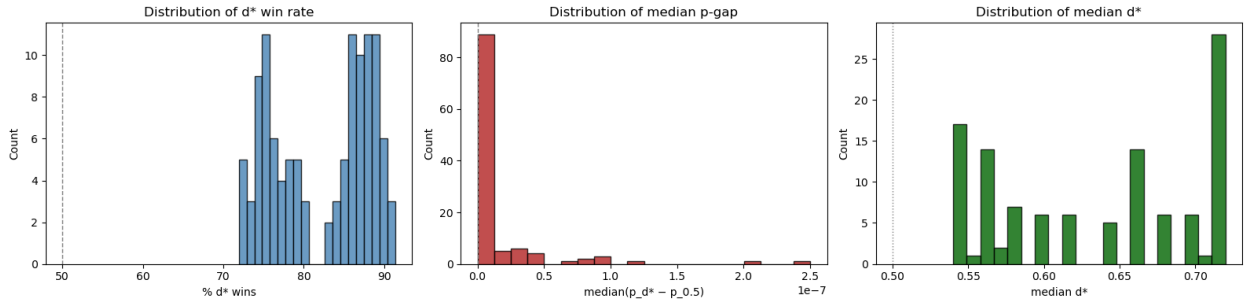


Figure 7: Distribution of monthly panel results across the September 2010 – February 2020 evaluation window. Left: histogram of the cross-sectional d^{LTM} win rate across evaluation dates. Centre: distribution of the median Ljung–Box p -value gap ($p^{(d^{\text{LTM}})} - p^{(0.5)}$) across dates. Right: distribution of the cross-sectional median d^{LTM} .

The win rate exceeds 50% at every single evaluation date across the entire decade, with the monthly distribution concentrated between 70% and 90%. This is not a result driven by a few exceptional months: the histogram in Figure 7 shows no mass below the 50% threshold, meaning d^{LTM} consistently outperforms fixed differencing regardless of the prevailing macro regime within the sample. The median Ljung–Box p -value gap is strictly positive throughout, and the cross-sectional median d^{LTM} ranges between 0.50 and 0.72, confirming that the evaluation window captures substantial and sustained deviation from the fixed benchmark. The Wilcoxon signed-rank test rejects $H_0 : p^{(d^{\text{LTM}})} = p^{(0.5)}$ at $p < 0.0001$ at every evaluation date.

8.5 Two-Date Cross-Sectional Analysis

To examine the cross-sectional structure of these wins, we select the two dates within the evaluation window where the median d^{LTM} departs most from 0.5 in opposite directions:

- **2010-09-09** — the post-GFC trough. Median $d^{LTM} = 0.58$, a gap of only 0.12 from the benchmark. This is the date on which the fixed- d assumption is most *defensible*: the cross-sectional distribution is closest to 0.5, and many tickers have d^{LTM} near the benchmark value. A strong d^{LTM} performance here therefore represents a demanding test.
- **2020-02-20** — the pre-COVID peak. Median $d^{LTM} = 0.72$, a gap of 0.22. This date captures the end of the longest sustained deviation from 0.5 in the sample, immediately before the COVID shock disrupts the persistence regime. It represents the date on which fixed differencing is most severely misspecified.

Together, these two dates bracket the performance of d^{LTM} under opposite conditions — low and high divergence from the benchmark — and test whether the advantage is concentrated only in extreme regimes or present more broadly.

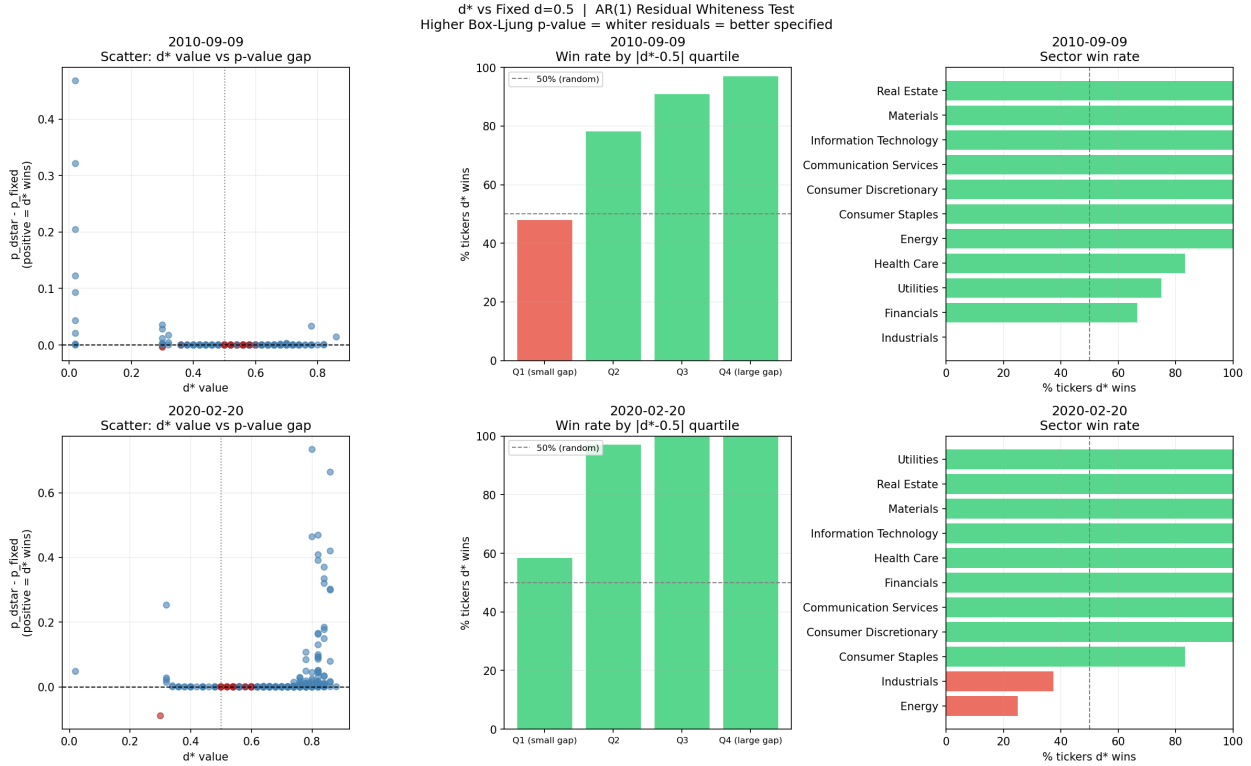


Figure 8: d^{LTM} vs fixed $d = 0.5$: AR(1) residual whiteness test at the two maximum-divergence dates. Left column: scatter of d^{LTM} value against Ljung–Box p -value gap (positive = d^{LTM} wins). Centre column: win rate by quartile of $|d^{LTM} - 0.5|$. Right column: sector-level win rates.

The results exhibit a clear dose-response pattern (Table 5): when d^{LTM} is close to 0.5 (Q1), the win rate is near 50% — essentially random, as expected when the two approaches agree. As $|d^{LTM} - 0.5|$ increases, the win rate rises monotonically, reaching 97–100% in Q4 at both dates. This is precisely the predicted behaviour if d^{LTM} correctly identifies the operative memory structure: the

benefit materialises exactly where it departs most from the fixed benchmark, and is negligible where the two approaches coincide.

Table 5: Win rate by quartile of $|d^{\text{LTM}} - 0.5|$. Q1 = smallest deviation; Q4 = largest. Wilcoxon signed-rank test yields $p < 0.0001$ at both dates.

Quartile	2010-09-09		2020-02-20	
	n	Win%	n	Win%
Q1 (small gap)	94	47.9%	101	58.4%
Q2	55	78.2%	102	97.1%
Q3	66	90.9%	108	100.0%
Q4 (large gap)	67	97.0%	89	100.0%
Overall	282	75.5%	400	88.8%

Notably, even at 2010-09-09 — the date most favourable to the fixed benchmark — the overall win rate of 75.5% is well above chance, and Q4 already achieves 97.0%. The higher win rate at 2020-02-20 (88.8% overall, 100% in Q3 and Q4) is consistent with the larger median gap: when fixed differencing is more severely misspecified, d^{LTM} wins more decisively.

The sector-level panels in Figure 8 confirm that the superiority of d^{LTM} holds broadly across sectors at both dates. The exceptions — Energy and Industrials at February 2020 — are economically interpretable: both sectors faced idiosyncratic disruption in the weeks immediately preceding the COVID-19 shock, conditions under which structural breaks may temporarily overwhelm a well-specified persistence estimate.

9 Conclusion and Future Work

This paper develops a finite-sample framework for long-memory estimation based on the detectability boundary of hyperbolic autocorrelation structure. Rather than targeting asymptotic identification of the fractional integration parameter, the procedure explicitly acknowledges finite-sample resolution limits and identifies the largest differencing order at which long-range dependence remains empirically distinguishable from noise. Monte Carlo analysis establishes the existence and scaling behavior of a detectability threshold, while empirical results on U.S. equities show that the resulting detectability boundary d_{LTM} , estimated in a walk-forward manner, produces materially whiter residuals than fixed fractional differencing across a full decade and broad cross-section of assets.

The empirical analysis further demonstrates that d_{LTM} behaves as a genuine state variable rather than a static parameter: it varies coherently over time, exhibits strong cross-sectional heterogeneity, and converges to stable values as sample size increases. These properties distinguish the detectability-based approach from classical frequency-domain estimators, which collapse to boundary solutions under realistic market conditions.

Several directions for future research emerge naturally from the detectability framework.

First, changes in the detectability boundary itself may carry predictive information. Exploratory analysis suggests that the time-series variation Δd_{LTM} is associated with shifts in short-run return dynamics, with abrupt changes in d_{LTM} coinciding with improved predictability of autoregressive components. Formalizing Δd_{LTM} as a shock indicator or state-transition signal, and evaluating its incremental forecasting power relative to conventional volatility- or momentum-based measures, is a promising avenue for further study.

Second, the detectability boundary appears to act as a natural regime separator. Preliminary evidence indicates that conditioning returns on d_{LTM} partitions the data into regimes with markedly different distributional properties, including degrees of normality and tail behavior. A systematic investigation of regime-conditional return distributions, including goodness-of-fit tests and stability over time, could clarify whether d_{LTM} provides a principled basis for persistence-based regime classification distinct from volatility-driven segmentation.

Third, the behavior of d_{LTM} as a function of data granularity warrants closer examination. While the present study focuses on daily data and estimation windows on the order of several thousand observations, applying the detectability procedure across different sampling frequencies and window lengths could reveal how memory resolution interacts with temporal aggregation. Such analysis would help determine whether detectability boundaries are scale-specific or exhibit stable structure across resolutions.

Fourth, extending the framework beyond the stationary ARFIMA null remains an important theoretical direction. Real-world time series often exhibit shocks, structural breaks, and nonlinear dynamics that violate standard long-memory assumptions. Studying the behavior of d_{LTM} under ARFIMA models augmented with jumps, regime switches, or heavy-tailed innovations, as well as applying the method to non-market datasets, would clarify the robustness and scope of the detectability concept outside idealized settings.

Fifth, a deeper theoretical characterization of the finite-sample resolution limit d_0 itself remains an important open problem. While the present paper establishes the existence and empirical scaling behavior of $d_0(T)$ via Monte Carlo calibration, a first-principles derivation based on the joint distribution of sample autocorrelations in the long-memory regime would substantially strengthen the theoretical foundation of the detectability framework. Deriving an explicit approximation for d_0 as a function of sample size, tail-window geometry, and noise variance—potentially via asymptotic expansions of log-ACF regression error or concentration bounds for tail statistics—would clarify the precise mechanisms governing memory resolvability and reduce reliance on simulation-based calibration.

Sixth, the detectability framework naturally motivates a systematic predictive comparison across forecasting models and trading strategies. While the present study focuses on residual whitening as a specification diagnostic, future work could evaluate the incremental forecasting and economic value of using d_{LTM} -based differencing within machine-learning models relative to fixed fractional differencing baselines. This includes comparing linear models, tree-based methods, and neural architectures trained on fractionally differenced series using either d_{LTM} or conventional choices (e.g. $d = 0.5$), as well as translating forecast improvements into explicit trading strategies with transaction costs and risk controls. Such analysis would help distinguish improvements in statistical specification from economically meaningful gains, and clarify when adaptive memory removal materially enhances real-world decision-making.

More broadly, the detectability-boundary perspective reframes long-memory estimation as a finite-sample identification problem rather than an asymptotic one. By targeting the point at which persistence becomes empirically resolvable, the approach delivers stable estimates and economically meaningful heterogeneity in environments where classical estimators fail. This shift in perspective opens new avenues for studying persistence as a dynamic, state-dependent feature of time series rather than a fixed structural constant.

Acknowledgments

I am grateful to Professor Dmitry Udler for valuable discussions and advice throughout the development of this work.

A Monte Carlo Performance Summary of d^{LTM} Estimator

Table 6 reports Monte Carlo performance statistics for the proposed tail-based procedure across a grid of true fractional integration parameters d_{true} . For each value, 200 independent ARFIMA paths are simulated. The column n_{ok} reports the number of paths for which a stable interior maximizer is obtained, while the failure rate reflects non-convergent or boundary outcomes. For values of $d_{\text{true}} < d_0$, the reported statistics summarize raw, non-convergent outputs and are included for diagnostic purposes only. Final calibrated estimates $\hat{d}$ used for inference are obtained via monotonic isotonic regression after entropy-based classification.

Table 6: Monte Carlo summary statistics for d^{LTM} across values of d_{true} .

d_{true}	n_{paths}	n_{ok}	Fail rate	Mean	Median	Bias	MAE	RMSE	$q_{0.10}$	$q_{0.90}$
0.10	200	106	0.470	0.000	0.00	-0.100	0.100	0.100	0.000	0.000
0.20	200	101	0.495	0.000	0.00	-0.200	0.200	0.200	0.000	0.000
0.24	200	101	0.495	0.000	0.00	-0.240	0.240	0.240	0.000	0.000
0.26	200	99	0.505	0.003	0.00	-0.257	0.257	0.257	0.000	0.000
0.28	200	97	0.515	0.020	0.00	-0.260	0.260	0.262	0.000	0.060
0.30	200	113	0.435	0.043	0.04	-0.257	0.257	0.260	0.000	0.100
0.40	200	176	0.120	0.131	0.14	-0.269	0.274	0.279	0.060	0.180
0.50	200	199	0.005	0.215	0.22	-0.285	0.285	0.292	0.136	0.280
0.60	200	200	0.000	0.334	0.34	-0.266	0.266	0.273	0.260	0.400
0.70	200	200	0.000	0.436	0.44	-0.264	0.264	0.272	0.340	0.520
0.80	200	200	0.000	0.579	0.58	-0.221	0.221	0.232	0.500	0.660
0.90	200	200	0.000	0.697	0.70	-0.203	0.203	0.217	0.600	0.800
1.00	200	199	0.005	0.813	0.84	-0.187	0.187	0.194	0.760	0.840
1.10	200	191	0.045	0.839	0.84	-0.261	0.261	0.261	0.840	0.840

For completeness, we verified that the reported Monte Carlo patterns persist on a finer grid with step size 0.02. The resulting median response and dispersion curves are smooth and monotone, with no additional qualitative features beyond those visible at the coarser resolution. Detailed fine-grid results are omitted for brevity.

We cap the search grid at $d = 0.84$, corresponding to $d^* = 1$ in the calibration, since larger values are not considered in the market setting analyzed here.

We emphasize that rows corresponding to $d_{\text{true}} < d_0$ do not represent meaningful estimates of long-term memory, as d^{LTM} is not well defined in this regime. These cases are retained solely to illustrate the finite-sample failure modes of tail-based objectives.

Table 7: Monte Carlo Performance of the Extended Whittle Estimator

d^*	n_{paths}	Bias	MAE	RMSE	Med($\hat{d}$)	Q ₁₀	Q ₉₀
0.00	200	0.021	0.021	0.037	0.000	0.000	0.080
0.10	200	−0.002	0.046	0.058	0.100	0.018	0.180
0.20	200	−0.012	0.048	0.061	0.200	0.100	0.260
0.30	200	−0.006	0.046	0.058	0.280	0.220	0.360
0.50	200	−0.006	0.046	0.058	0.500	0.420	0.580
0.70	200	−0.002	0.046	0.060	0.700	0.620	0.780
0.90	200	0.001	0.045	0.057	0.900	0.840	0.980
1.10	200	0.007	0.046	0.059	1.100	1.020	1.180
1.30	200	0.002	0.037	0.045	1.320	1.238	1.340

Note. Each row summarizes 200 Monte Carlo simulations of ARFIMA(0, d^* , 0) processes. The table reports finite-sample performance of the *extended Whittle estimator* under the same experimental design used throughout the paper. Bias, MAE, and RMSE are computed relative to the true memory parameter d^* , while quantiles summarize estimator dispersion across simulation paths.

A.1 Extended Whittle Estimates on Market Data

This appendix reports the empirical behavior of the extended Whittle (EW) estimator when applied to the full universe of equity log-price series used in the main text. For each asset, the EW memory parameter was estimated on the same sample used for the tail-based procedure.

Table 8 summarizes the cross-sectional distribution of EW estimates. Despite strong Monte Carlo performance under correctly specified ARFIMA dynamics (see Appendix 7), EW estimates on market data collapse near the nonstationary boundary. In particular, more than 80% of assets receive estimates in the range $[0.98, 1.02]$, with a median estimate equal to 1.0 and no meaningful cross-sectional variation.

These results indicate that, in empirical financial settings, frequency-domain likelihood methods such as EW are unable to resolve economically relevant differences in long-term dependence across assets.

Table 8: Cross-Sectional Summary of Extended Whittle Estimates on Market Data

Statistic	Value
Number of assets	484
Mean of $\hat{d}_{\text{EW}}$	0.988
Standard deviation	0.021
Minimum	0.880
25% quantile	0.980
Median	1.000
75% quantile	1.000
Maximum	1.000
Assets with $\hat{d}_{\text{EW}} \in [0.98, 1.02]$	406 (83.9%)

A.2 Entropy diagnostics for detectability classification

Table 9: Entropy diagnostics for the tail–slope curve across values of d_{true} . Reported values are Monte Carlo averages computed over successful replications.

d_{true}	Mean global entropy	Mean local entropy	Entropy ratio	n_{ok}
0.00	0.7425	0.6015	0.8100	198
0.05	0.7493	0.6171	0.8235	198
0.10	0.7617	0.6470	0.8495	193
0.15	0.7517	0.6307	0.8391	195
0.20	0.7394	0.6015	0.8135	197
0.25	0.7243	0.5909	0.8159	199
0.30	0.7172	0.5268	0.7346	199
0.35	0.7122	0.5086	0.7141	200
0.40	0.7058	0.4332	0.6138	199
0.45	0.6756	0.3384	0.5009	200
0.50	0.6602	0.3108	0.4708	200
0.55	0.6319	0.2622	0.4150	200
0.60	0.6090	0.2827	0.4642	200
0.65	0.5629	0.2619	0.4653	200
0.70	0.5270	0.2646	0.5021	200
0.75	0.4903	0.2715	0.5538	200
0.80	0.4617	0.3017	0.6534	199
0.85	0.4308	0.3442	0.7989	199
0.90	0.4097	0.4337	1.0585	200
0.95	0.3703	0.4925	1.3300	200
1.00	0.3589	0.4928	1.3728	200

Consistent with the detectability framework, entropy is high when the true memory parameter satisfies $d^* < d_0$ and decreases sharply in the resolvable long–memory regime. At very large values of d^* , the slope–entropy can rise again, reflecting increased variability of tail–slope estimation at the extreme end of the parameter range in finite samples.

We emphasize that the observed entropy patterns are finite–sample phenomena documented in Monte Carlo calibration and are not claimed as universal asymptotic properties of fractionally integrated processes.

B Auxiliary Classification Results for Detectability Filtering

To separate cases where the tail–based detectability index d^{LTM} is non–informative from those exhibiting genuine long–term memory structure, we train a binary classifier using entropy–based diagnostics and related features. The binary label is defined as

$$y = \begin{cases} 0, & \text{false } d^{\text{LTM}} \quad (d^* < 0.3), \\ 1, & \text{true } d^{\text{LTM}}. \end{cases}$$

The classifier is trained on a dataset of $N = 1252$ observations constructed from entropy, mutual information, and Hurst–type features. Table 10 reports standard classification metrics, while Table 11 shows the corresponding confusion matrix.

Table 10: Classification report for entropy-based detectability classifier.

Class	Precision	Recall	F1-score	Support
0 (false d^{LTM})	0.891	0.997	0.941	351
1 (true d^{LTM})	0.999	0.952	0.975	901
Accuracy	0.965			
Macro average	Precision = 0.945, Recall = 0.975, F1 = 0.958			
Weighted avg	Precision = 0.968, Recall = 0.965, F1 = 0.965			

Table 11: Confusion matrix for entropy-based detectability classifier.

	Predicted 0	Predicted 1
True 0	350	1
True 1	43	858

The resulting classifier achieves an area under the ROC curve of

$$\text{AUC} = 0.9985,$$

indicating near-perfect separability between false and true detectability cases. The trained model is saved for reproducibility and downstream filtering of non-informative d^{LTM} estimates.

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